

Representation Genesis: Wisdom-Oriented Agents Invent Better Languages for Future Thought

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Abstract

AI systems are usually optimized to produce better answers, policies, or single-shot benchmark scores. We study a complementary objective: producing better representations for future thought. We call this problem *Representation Genesis*. In this framework, macros, ontologies, theorem templates, task schemas, skill APIs, and evidence gates are treated as evolvable objects. A representation is valuable only if it compresses future problems after paying its unpacking cost, transfers across tasks, remains robust under perturbation, and preserves provenance. We introduce a representation score combining compression gain, unpacking cost, reuse, transfer, robustness, novelty, and residual absorption. We report the first reproducible artifacts: a toy monoid macro-search lab, a 130-file Wisdom Science macro-mining pass, a 10-claim machine-readable registry, and generated LaTeX tables used by this draft. We further introduce *Perspectival Grounding*: the view is not the world, so a wise representation must preserve competing interpretations and select evidence-gathering views that collapse ambiguity without prematurely collapsing truth.

1 Introduction

Current AI evaluation asks whether a model can answer the present problem. Wisdom Science asks a different question: after experience, failure, and feedback, does the system become better equipped for future problems? Representation Genesis is the upstream version of this question. Instead of only changing answers, a wise system changes the language in which future answers become shorter, safer, and more transferable.

Thesis. Wisdom-oriented agents should search over representations, not only over outputs. The relevant product of a learning system is sometimes a macro, task schema, proof route, failure ontology, or API boundary that makes many future tasks easier.

Boundary. This paper does not claim that compression, macro discovery, or library learning are new. The contribution is narrower: a governance framework for representation search inside a wisdom-oriented agent, with explicit evidence gates and transfer-oriented metrics.

2 Motivation

The Wisdom Science corpus already contains repeated conceptual structure: WisdomBench, Cognitive Immunity, Evidence Gates, the Intelligence-Wisdom Gap, and the Second Scaling Law. This repetition is not merely stylistic. It means that the research program has developed a reusable language. If that language is left implicit, papers drift, formulas fork, and claims become hard to audit. If it is made explicit, the same language becomes a compression layer for future research.

3 Problem Formulation

Definition 1 (Representation Object). *A representation object r is any reusable structure that changes how future tasks are encoded, solved, or audited. Examples include macros, ontologies, theorem templates, task schemas, skill APIs, prompt schemas, failure taxonomies, and evidence gates.*

Definition 2 (Representation Genesis). *Representation Genesis is the process of proposing, evaluating, retaining, and revising representation objects under compression, transfer, robustness, novelty, and provenance constraints.*

Definition 3 (Representation Score). *For a representation r evaluated on corpus $\mathcal{C}$ and held-out tasks $\mathcal{H}$, define*

$$S(r) = \lambda_c G_c(r) - \lambda_u U(r) + \lambda_\rho R(r) + \lambda_\tau T(r) + \lambda_h H(r) + \lambda_n N(r) + \lambda_a A(r),$$

where G_c is compression gain, U is unpacking cost, R is reuse, T is transfer gain, H is robustness, N is novelty distance, and A is residual absorption.

Compression gain. A representation must shorten a corpus after paying definition cost:

$$G_c(r) = L(\mathcal{C}) - [L(r) + L(\mathcal{C} \mid r)].$$

Unpacking cost. High compression is not useful if the representation becomes too expensive to explain or verify.

Transfer gain. A representation should improve held-out tasks, not merely memorize the source corpus.

Residual absorption. Failures are especially valuable when they expose parts of the problem space that the current representation cannot compress.

4 Algorithmic Loop

Representation Genesis follows a simple loop:

1. **Propose:** generate a candidate macro, schema, ontology, or evidence rule.
2. **Pack:** measure whether it compresses a source corpus after paying definition cost.
3. **Unpack:** test whether another system can recover the intended meaning.
4. **Transfer:** evaluate held-out tasks, papers, or code modules.
5. **Stress:** perturb wording, task distribution, or evidence source.
6. **Retain:** keep a Pareto frontier rather than a single winner.

Proposition 1 (Why a Pareto frontier is necessary). *If representations are scored only by raw compression, then the optimum may hide assumptions or destroy transfer. Therefore a wisdom-oriented representation archive must preserve non-dominated candidates across compression, unpacking cost, transfer, robustness, novelty, and provenance.*

Table 1: Top seed terms from the Wisdom Science corpus. Counts and document frequencies are generated from `macro_registry_v0.json`.

Term	Count	Doc. freq.
immunity	502	33
WisdomBench	178	24
Cognitive Immunity	144	27
Ouroboros	106	31
homeostasis	75	19
plasticity	64	14
Second Scaling Law	52	13
Wisdom Quotient	49	22
Cognitive Entropy	40	9
Intelligence-Wisdom Gap	35	12
Evidence Gate	32	13
WisdomBench-Embodied	26	10

5 Reproducible Artifacts

We implement the first version as dependency-free scripts:

- `experiments/representation_lab_toy_monoid.py`;
- `experiments/representation_lab_metrics.py`;
- `experiments/wisdom_science_macro_mining.py`;
- `experiments/wisdom_science_claim_registry.py`;
- `experiments/generate_wisdom_science_tables.py`.

Toy monoid. The toy lab verifies that macros compress only real source substrings. It explicitly avoids the failure mode in which generated macro symbols are recursively compressed as if they were new source structure. The current demo compresses a 53-symbol corpus to 48 symbols after definition cost, yielding a compression ratio of 1.10 with one real macro.

Corpus macro mining. The macro-mining pass scans 130 paper and code files. The strongest seed terms are shown in Table 1.

Representation-compression study. We also run a deterministic macro-rewriting experiment on 12 core Wisdom Science source files. The canonical macro glossary contains 13 long-form terms such as Wisdom Quotient, Second Scaling Law, Cognitive Immunity, WisdomBench-Embodied, and Representation Genesis. After paying the glossary cost, the macro-coded corpus saves 3079 bytes and yields a compression ratio of 1.0126. The result is intentionally modest: it is evidence that a shared representation language has measurable compression value, not evidence that compression alone is wisdom.

Claim registry. We export the core claims into JSON, CSV, Markdown, and LaTeX tables. This prevents a common failure mode in ambitious research programs: claims become stronger as they move from notes to papers. The registry records claim, evidence source, strength, and limitation.

Table 2: Canonical macro counts in the representation-compression study. Generated from `representation_compression_v0.json`.

Macro	Long form	Count
WB	WisdomBench	50
CI	Cognitive Immunity	30
SSL	Second Scaling Law	27
EG	Evidence Gate	25
RG	Representation Genesis	20
IWG	Intelligence-Wisdom Gap	18
WB-E	WisdomBench-Embodied	17
WQ	Wisdom Quotient	16
EFI	Embodied Failure Immunity	11
COS	Cognitive Operating System	5

6 Perspectival Grounding

Representation Genesis is not only about shorter language. It is also about choosing a language that does not confuse a projection with the world. A glyph that appears to be “16” may become “91” under rotation or may belong to another writing system. A Chinese character may carry multiple senses, emotions, and social uses until collocation and discourse context are observed. An occluded object may have several affordances, and a strategic signal may be literal, a bluff, or a second-order feint.

Definition 4 (Perspectival Grounding). *Let x be a latent state, v a viewpoint or context, and $o = \pi_v(x)$ an observation. Perspectival Grounding is the discipline of preserving the hypothesis set*

$$\mathcal{H}(o) = \{x' : \pi_v(x') = o\}$$

until additional views, actions, context, or social evidence reduce ambiguity.

This turns ambiguity from a nuisance into a representation objective. The agent should not merely answer from the first projection; it should ask which next view, touch, query, or belief-model update has the best information gain per cost.

The current dependency-free lab contains six cases across glyphs, language, robotics, deictic social instruction, and counter-reflexive game signals. The mean initial ambiguity is 1.56 bits, the mean final ambiguity is 0 bits, and every case is resolved after evidence-gathering views are added. This is not a claim that the toy lab solves perception. It is a guardrail: any future wisdom representation must distinguish surface observation, latent state, viewpoint, social context, and evidence cost.

7 Evidence Path

Representation Genesis is designed as a toy-to-real evidence chain:

1. **Toy monoid lab:** verify that proposed macros compress after definition cost and do not recursively compress generated symbols.
2. **Research corpus:** mine the Zenodo/P0–P7 corpus for repeated claims, formulas, evidence gates, and terminology drift.

Table 3: Machine-generated claim registry used to keep Wisdom Science claims auditable.

ID	Claim	Evidence	Strength
C1	Wisdom should be distinguished from first-round intelligence.	P2, P3	strong
C2	Wisdom can be operationalized as normalized cross-round learning gain.	P1, P2	strong
C3	Higher first-round intelligence does not guarantee higher wisdom.	P3, P4	medium-strong
C4	Wisdom scales with architectural qualities and experience, not only parameters.	P4	medium
C5	Failures can be converted into transferable immunity signals.	P1	medium-strong
C6	Embodied agents should report longitudinal learning ability.	P5, P6, P7	strong
C7	Robot learning ability can be described by embodied plasticity, immunity, and homeostasis.	P5, P6	medium
C8	Evidence gates are necessary for embodied wisdom claims.	P7	strong
C9	Representation search is the next layer of wisdom-oriented systems.	P8 draft	new
C10	Embodied failure trajectories can become transferable antigens.	P9 draft	new
C11	A single observation should be treated as a projection rather than the latent world.	P8 toy lab	new
C12	Embodied wisdom is distributed across organ-like subsystems, not only a brain policy.	P9 schema	new

Table 4: Perspectival Grounding toy lab. Each case starts from a single ambiguous projection and chooses low-cost additional evidence until the latent hypothesis set is resolved. Generated from `perspectival_grounding_v0.json`.

Case	Domain	$ \mathcal{H}_0 $	H_0	Drop	Cost
rotated 16 91	glyph	3	1.58	1.58	1.0
six nine orientation	glyph	2	1.00	1.00	0.5
polysemous chinese xing	language	4	2.00	2.00	0.5
occluded mug can bowl	robotics	3	1.58	1.58	1.5
deictic instruction	social_robotics	3	1.58	1.58	0.8
counter reflexive game	social_game	3	1.58	1.58	1.2
Mean	–	–	1.56	1.56	–

3. **Code corpus:** mine SOVEREIGN modules for reusable skill and pipeline abstractions.
4. **Benchmark schemas:** rewrite WisdomBench and WB-E task schemas using the discovered representation language, then measure prompt cost and consistency.
5. **Perspectival grounding:** test whether ambiguous observations retain multiple hypotheses until active views, context, or counter-reflexive belief models resolve them.

8 Relation to Wisdom Science

Representation Genesis sits above the existing Wisdom Science stack. WisdomBench measures whether agents improve across repeated experience. Cognitive Immunity studies whether failures become transferable rules. The Second Scaling Law studies how wisdom depends on architecture. Representation Genesis asks a more upstream question: can a system invent the language that makes all of these future measurements easier to express, verify, and extend?

9 Related Work

Compression and mathematics. Freedman et al. frame mathematical abstraction as compression over formal structure. Our work uses this as motivation but does not claim priority over compression accounts of mathematics.

Library learning and program synthesis. DreamCoder, Stitch, and LILO learn reusable program abstractions. Representation Genesis differs by treating papers, benchmarks, evidence gates, failure taxonomies, and skill APIs as representation objects, not only programs.

Compression and intelligence. Schmidhuber and Hutter connect compression, curiosity, and universal intelligence. Our claim is not that compression is intelligence, but that representation governance is one measurable component of wisdom-oriented systems.

Symbolic law discovery. AI Feynman and symbolic-regression systems search compact laws. Representation Genesis searches a broader class of research language: formulas, macros, schemas, and provenance rules.

Active perception and symbol grounding. Active perception and next-best-view robotics study how agents gather information under partial observability, while symbol grounding studies how signs acquire meaning through perception and action. Perspectival Grounding does not claim priority over those fields. It contributes a Wisdom Science formulation: ambiguity, viewpoint, semantic context, and counter-reflexive belief should be reported as evidence objects rather than hidden inside a single model answer.

10 Limitations

Compression is not truth. A shorter representation may hide assumptions, overfit to a corpus, or reduce interpretability. The current macro miner is lexical rather than semantic. The current toy monoid is a sanity check, not evidence about real mathematics. Future work must test whether representation changes improve held-out writing, benchmark design, and embodied failure transfer. The Perspectival Grounding lab is intentionally toy-scale; real use requires multimodal perception logs, active sensing traces, social context, and adversarial or game-theoretic evaluation.

11 Conclusion

Representation Genesis treats better language as a first-class product of intelligent systems. If WisdomBench measures whether agents learn from experience, Representation Genesis asks whether they can invent the representations that make future learning easier.

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